



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

ATHEROSCLEROSIS ISCHEMIC HEART DISEASE

CARDIOVASCULAR SYSTEM PPOM3

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Session Objectives

- 1. Define **atherosclerosis** and **atheroma** and describe the **three major components** of an **atheroma**.
- 2. Discuss and describe the role of **endothelial injury** and **inflammation** in the **pathogenesis** of **atherosclerosis**.
- 3. Discuss and describe the **role of hypercholesterolemia** and **hemodynamic effects** in the **pathogenesis** of **atherosclerosis**.
- 4. Discuss, describe, and compare **vulnerable** and **stable (mature) atherosclerotic plaques** and the **consequences** of **plaque formation** for each.
- 5. Describe and **define coronary artery disease, ischemic heart disease, the different types of angina** and **coronary artery thrombosis**.

OUTLINE/OVERVIEW

- Arteriosclerosis
- Atherosclerosis
- Atheroma/plaque
- Risk factors: modifiable and non-modifiable
- Atherogenesis
- Atherosclerosis pathogenesis
- Atherosclerosis progression
- Stable v. Vulnerable plaque
- Complications
- Coronary artery disease
- Ischemic Heart Disease/ Angina Pectoris

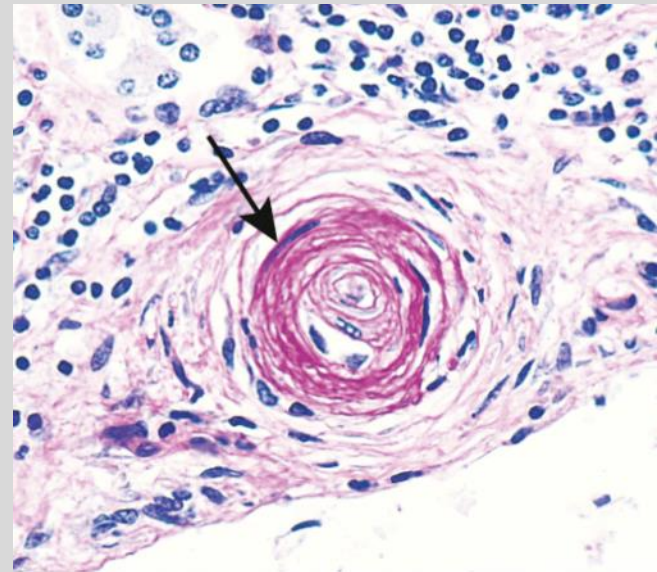
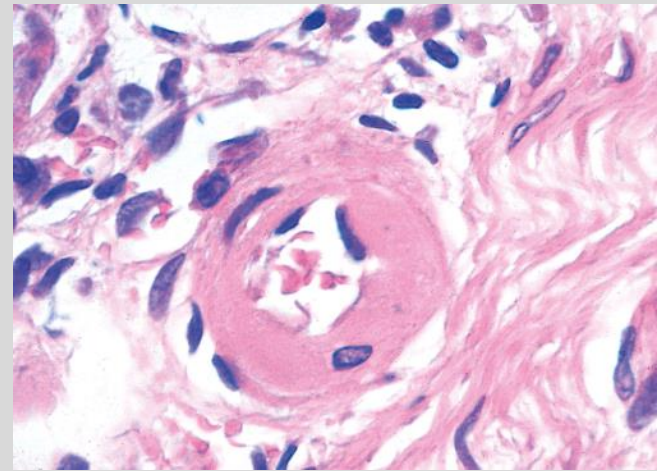
Arteriosclerosis

- “**hard arteries**”
 - Arterial wall thickening
 - Loss of elasticity
- Four general patterns
 - **Arteriolosclerosis** (small arteries and arterioles)
 - **Mönckeberg medial sclerosis** (affects the medial layer)
 - **Fibromuscular intimal hyperplasia** (muscular large arteries)
 - **Atherosclerosis** (intimal thickening)



Arteriolosclerosis

- Narrowing of **small arterioles**
- **Hyaline** arteriolosclerosis
 - Protein leaks into the wall
 - Common in essential HTN and Type II DM
- **Hyperplastic** arteriolosclerosis
 - Thickening of vessel wall by hyperplasia of smooth muscle
 - Results from **malignant HTN**
 - “Onion-skinning” pattern on histology

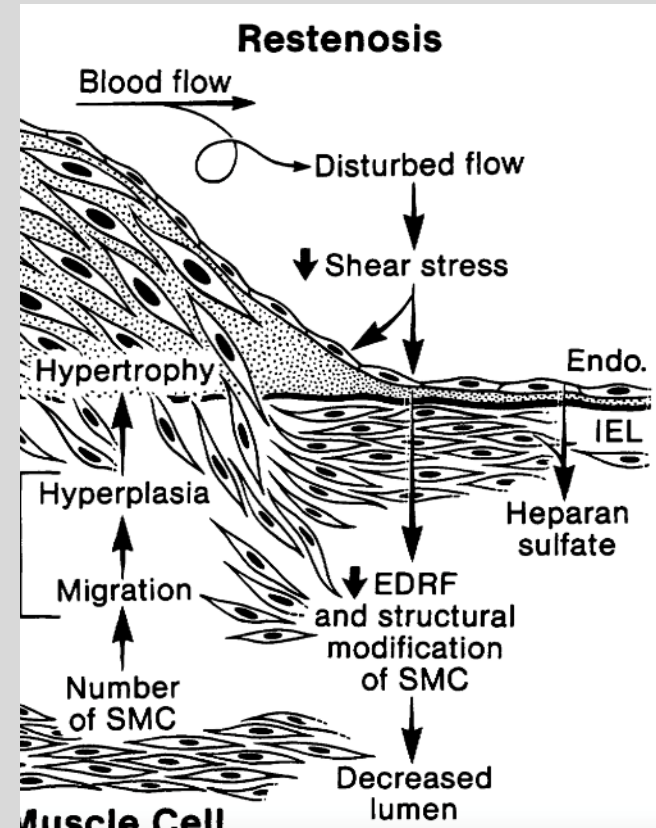


- **Calcification of the media of muscular arteries**
- **Non-obstructive**, not clinically significant
- “Pipestem” appearance on x-ray
- Can be seen on mammogram (calcification in a vascular pattern)



Fibromuscular Intimal Hyperplasia

- Driven by **inflammation or mechanical injury**
- Healing response leads to stenosis of the vessel
- Commonly seen in **stent restenosis**
- Note: **this is different from fibromuscular dysplasia**
 - A non-inflammatory, non-atherosclerotic vascular disease



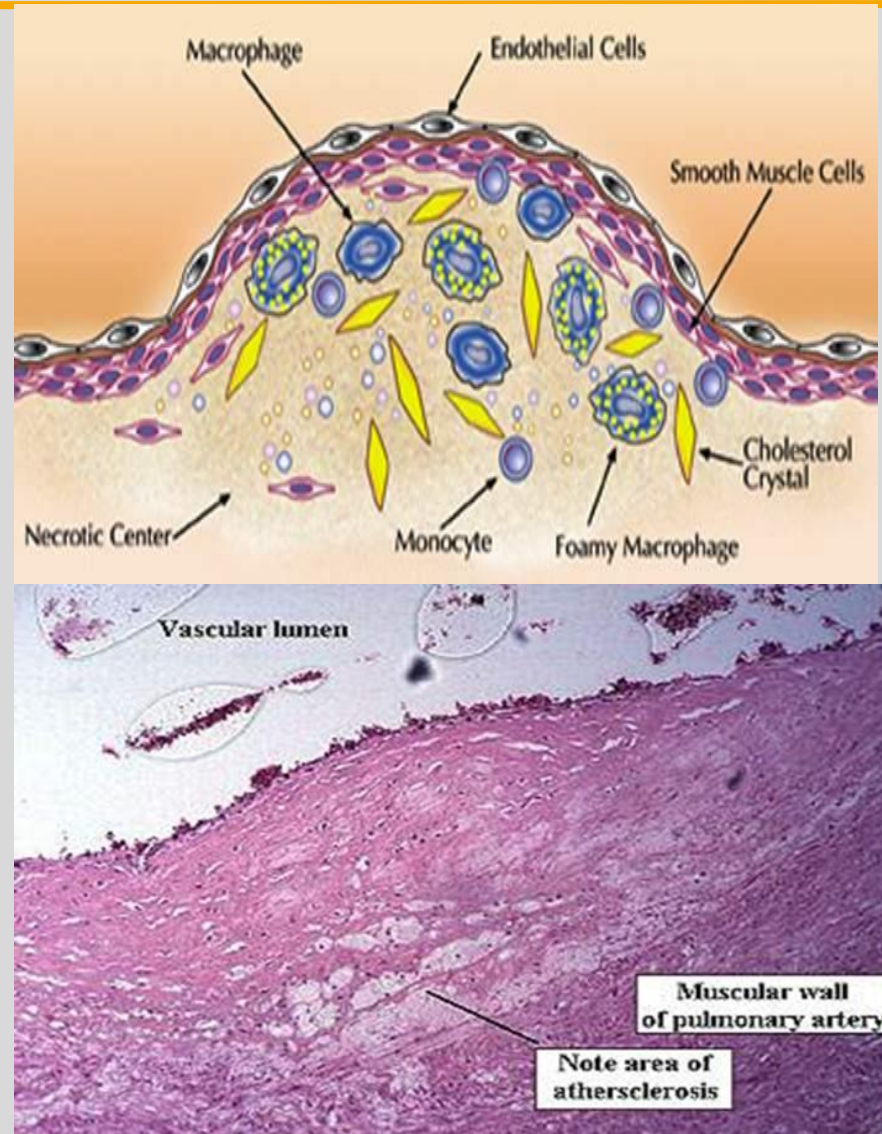
ATHEROSCLEROSIS



- Underlying pathology responsible for coronary, cerebral and peripheral vascular disease (arteries affected)
- Most commonly: **abdominal aorta, coronary arteries, popliteal arteries, internal carotid arteries, and the circle of Willis**
- Intimal thickening/plaque **obstructs blood flow (narrowing or occlusion)**
- **Media weakened**
- **Intimal lesions called atheromas** (atheromatous or atherosclerotic plaques) that protrude into the vessel lumen
- **INTIMAL THICKENING AND LIPID ACCUMULATION**

ATHEROMA

- RAISED FOCAL LESION STARTING WITHIN THE INTIMA
- SOFT, YELLOW, GRUMOUS CORE OF LIPID (CHOLESTEROL AND CHOLESTEROL ESTERS), COVERED BY FIBROUS CAP



ATHEROMA

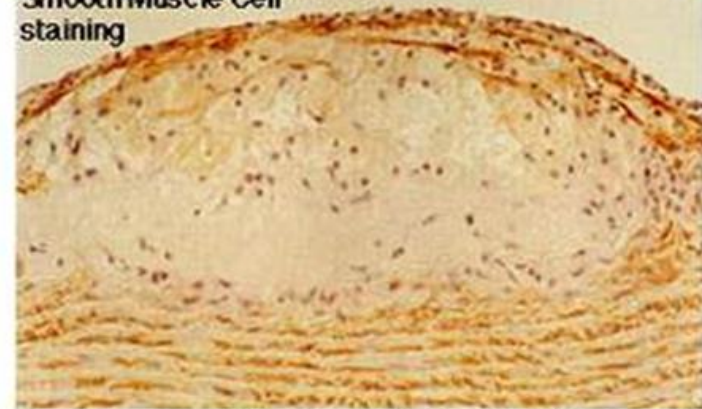
- Three major components:
 - - **Cells**: smooth muscle, macrophages, leukocytes (T lymphocytes)
 - - **ECM**: collagen, elastic fibers, proteoglycans
 - - Intracellular and extracellular **lipid**

Atherosclerotic Lesion - thoracic aorta

Macrophage staining



Smooth Muscle Cell staining



The Macrophage Home Page <http://www.path.ox.ac.uk/sg>

Atherosclerosis Risk Factors

- **Modifiable**
 - Smoking
 - Hypertension
 - Hyperlipidemia (**LDL increases risk, HDL decreases risk**)
 - Diabetes
 - Systemic Inflammation
 - C-reactive protein (CRP) is an acute phase reactant from the liver (augments immune response by binding to bacteria and activating complement cascade)
 - **CRP is an independent marker of risk for MI, stroke, PAD, and sudden cardiac death**
 - Limited evidence for CMV, Chlamydia, coxsackie B and H. pylori involved in pathogenesis

Atherosclerosis Risk Factors

- **Nonmodifiable**
 - Age (>60 years old)
 - Sex (**increased in men and postmenopausal women**)
 - Estrogen is protective in pre-menopausal women
 - **Family history** – most important independent risk factor
 - Genetic abnormalities
 - LDL receptor gene mutations
 - **Homocystinuria** – inborn error of metabolism causes elevated serum homocysteine levels (hyperhomocystinemia)
 - Associated with premature vascular disease (coronary atherosclerosis, PVD, stroke, and venous thrombosis)

Current model: **response to injury** – chronic **inflammatory** and healing response of the arterial wall initiated by **endothelial injury**

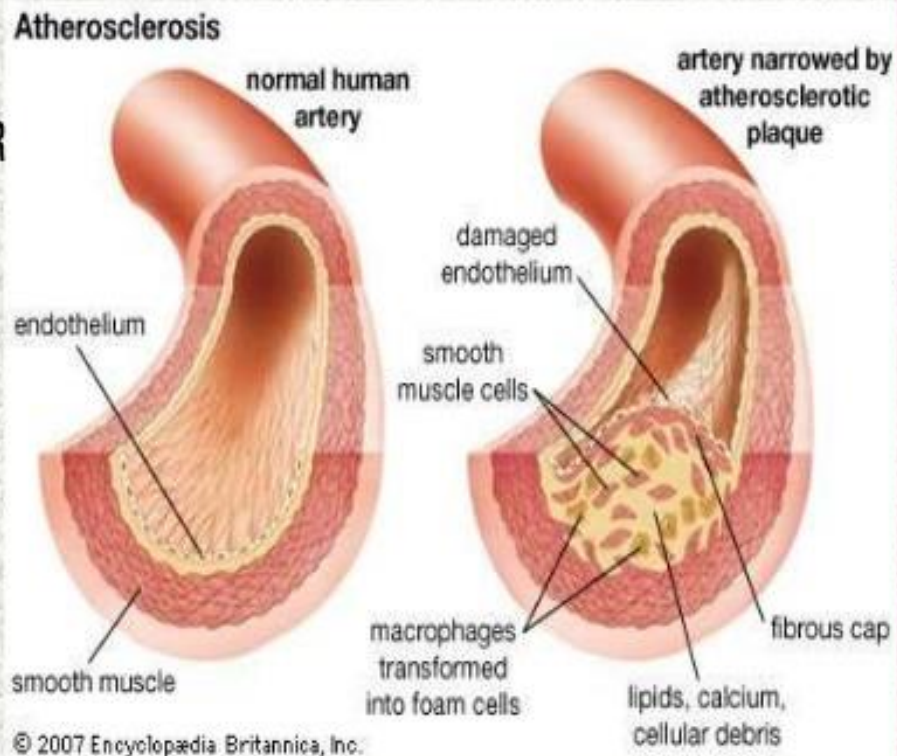
IMPORTANT CAUSES OF ENDOTHELIAL INJURY:

- **Hypercholesterolemia** - impairs endothelial cell function by increasing local ROS (reactive oxygen species)
- **Hemodynamic effects:**
 - Areas of disturbed, turbulent blood flow - more plaques
 - Laminar blood flow protects against AS
 - Hypertension directly increases hemodynamic shear stress on endothelial cells
- **Inflammation** – triggered by accumulation of cholesterol crystals and free fatty acids in macrophages and other cells
- **Chronic diseases: Hypertension, diabetes, obesity, etc.**

RESPONSE-TO-INJURY HYPOTHESIS

1) CHRONIC ENDOTHELIAL INJURY

- Hyperlipidemia
- Hypertension
- Smoking
- Homocysteine
- Hemodynamic factors
- Toxins
- Viruses
- Immune Reactions

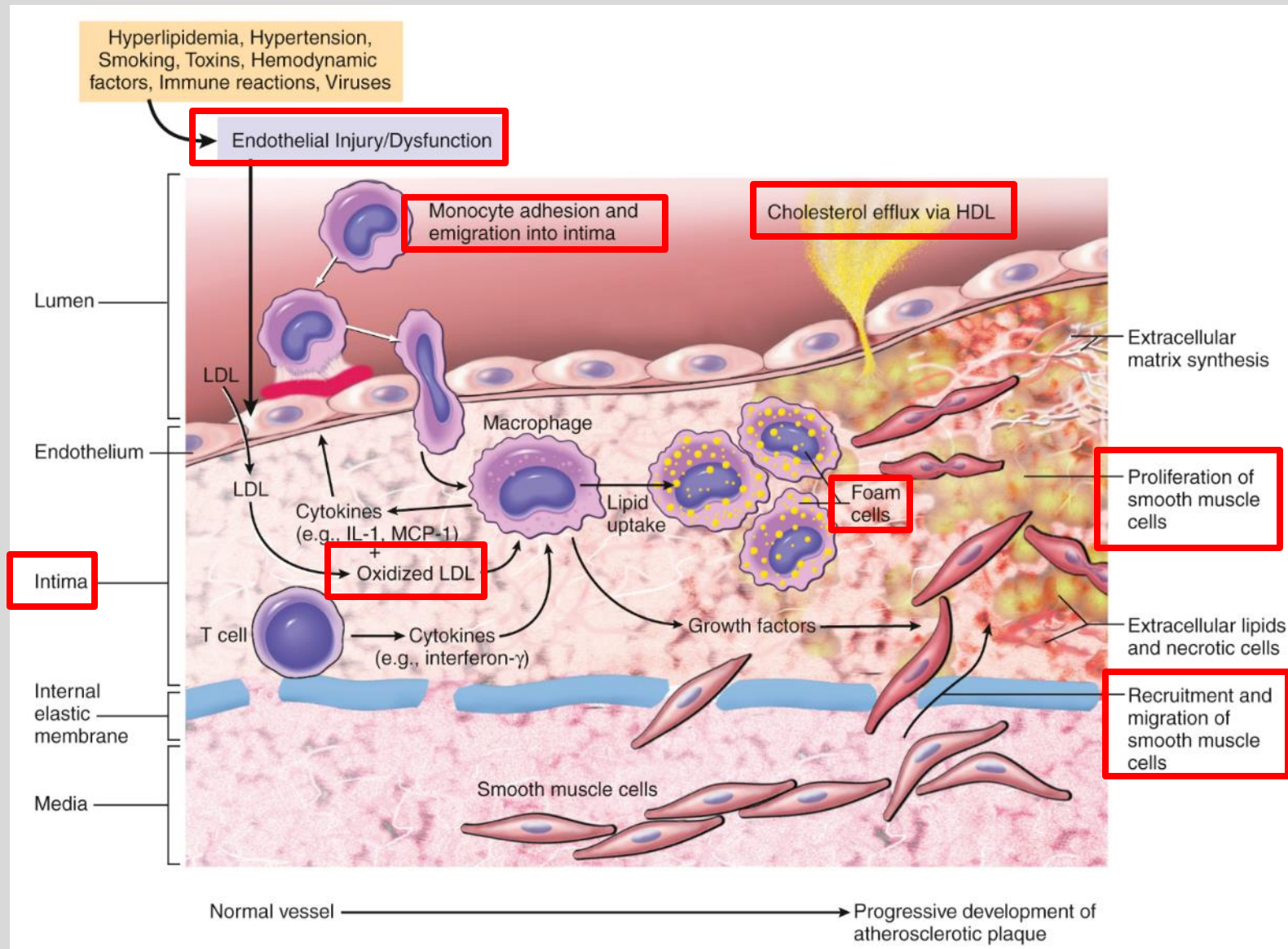


- LDL is mostly cholesterol
- Chronic hypercholesteremia may cause endothelial dysfunction by producing oxygen free radicals
- Lipoproteins accumulate within the intima
- LDL in macrophages oxidized by free radical generation and then engulfed by macrophages via a scavenger receptor (different than the LDL receptor)

Inflammatory cells in AS

- Inflammation triggered by accumulation of cholesterol crystals and free fatty acids in macrophages
- Dysfunctional endothelial cells express surface selective adhesion molecules, VCAM-1 -- binds monocytes and T lymphocytes
- **Monocytes adhere to endothelium, migrate to intima, and transform to macrophages that engulf oxidized LDL**
- Progressive accumulation of lipids
- Macrophages produce IL-1, TNF and chemokines further increasing adhesion of wbc's
- T-lymphs and monocytes also interact with cytokines and a **chronic inflammatory state develops**

Development of an Atheroma

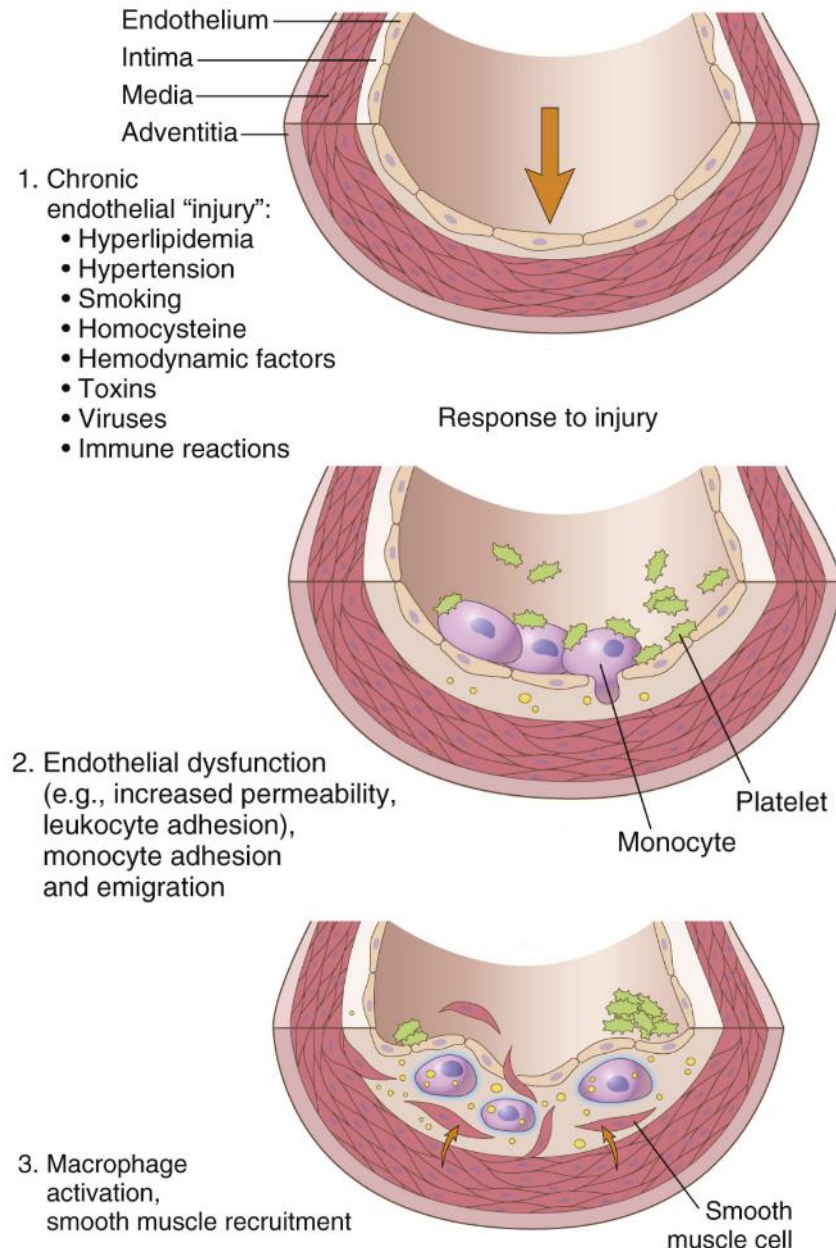


PATHOGENESIS: overview of progression

- **Endothelial injury: endothelial dysfunction** causing increased permeability, wbc adhesion, and exposure to thrombogenic agents (thrombosis)
- **Accumulation of lipoproteins (mostly LDL)**, in vessel wall; lipoproteins undergo oxidation; intracellular and extracellular accumulation of lipids
- **Monocytes (mostly) adhere to endothelium**, migrate into intima and transform into **macrophages/foam cells**; platelets also adhere to endothelium
- Factors released from activated platelets, macrophages, and vascular cells cause **smooth muscle cells to migrate from media to intima** (include PDGF and FGF)
- **Smooth muscle cells proliferate in intima and produce ECM so that collagen and proteoglycans accumulate and plaque matures**; also recruitment of T cells
- Lipid accumulation – extracellular and within macrophages and smooth muscle cells
- **Calcification of ECM and necrotic debris late in pathogenesis**

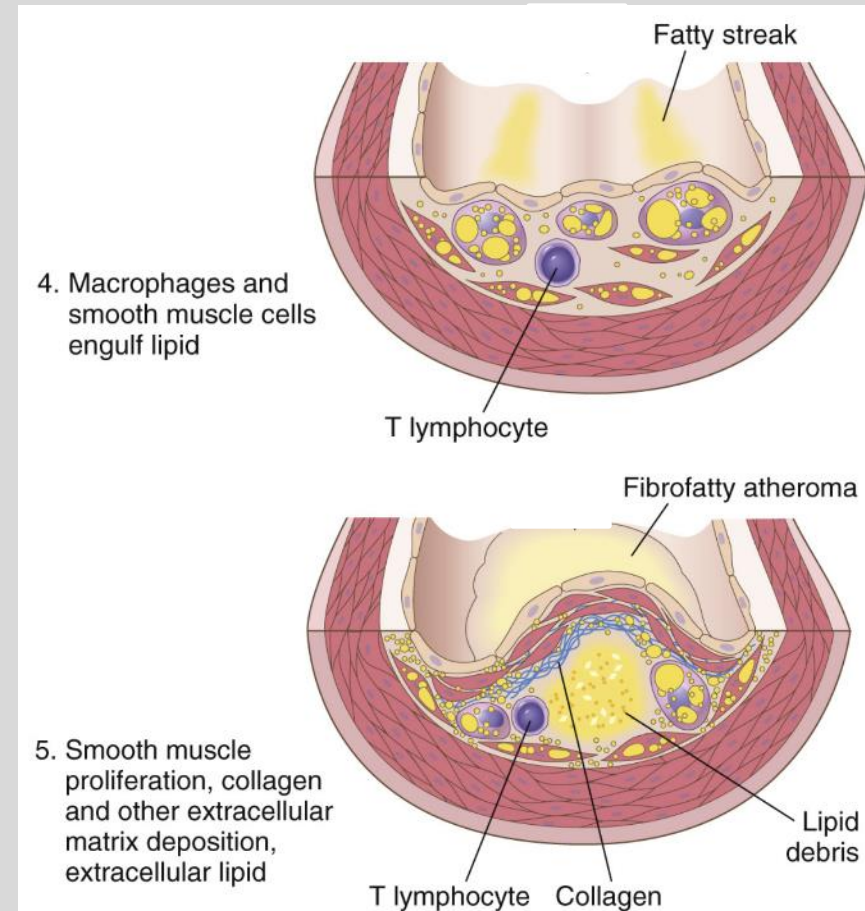
Pathogenesis

- 1. Endothelium damage** due to modifiable and nonmodifiable risk factors
- 2. Lipids (cholesterol) leak into the intima** (mainly LDL) due to endothelial dysfunction
- 3. Macrophages are activated and vascular smooth muscle cells move to the intima**
 - VSMC recruitment mediated by PDGF and FGF



Pathogenesis

4. **Lipid is oxidized**, and consumed by macrophages using the scavenger receptor
 - Macrophages become **foam cells**, creating the **fatty streak**
 - Platelet adhesion
5. **VSMC proliferation** and formation of the atheroma
 - Calcification of the ECM and necrotic debris late in pathogenesis



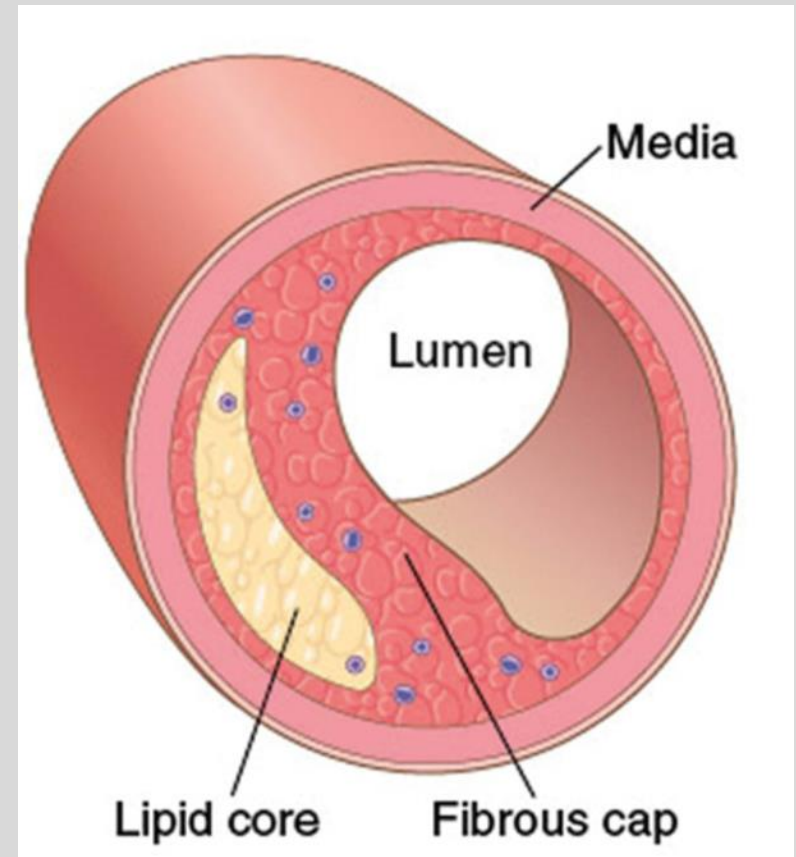
Fibrous Cap

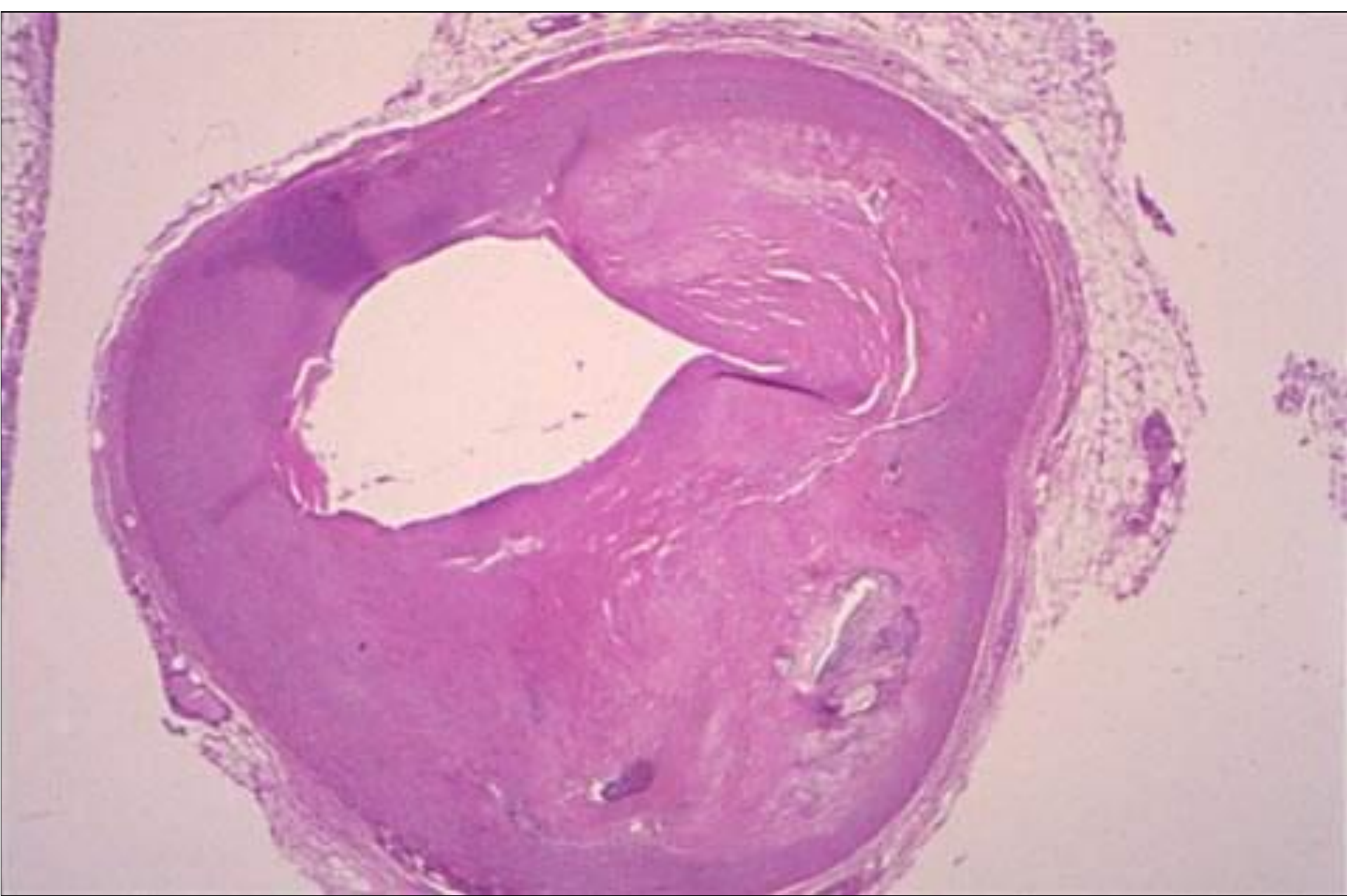
- Major structural component of fibrous cap is **dense collagen**
 - Establishes its strength and determines plaque stability
- Fibrous cap is constantly undergoing **remodeling**
 - **Changes the stability of the plaque**
- Collagen is made by smooth muscle cells and degraded by metalloproteinases
 - Enzymes from macrophages in atheroma
- Inflammation destabilizes plaque integrity

Maturation/stabilization of plaque

- Smooth muscle cells migrate from media to intima, proliferate, and elaborate ECM components
- Growth factors such as PDGF, FGF, and TGF contribute to smooth muscle cell proliferation
- Smooth muscle cells in vessel wall make extracellular matrix molecules (esp. collagen) which stabilize plaques

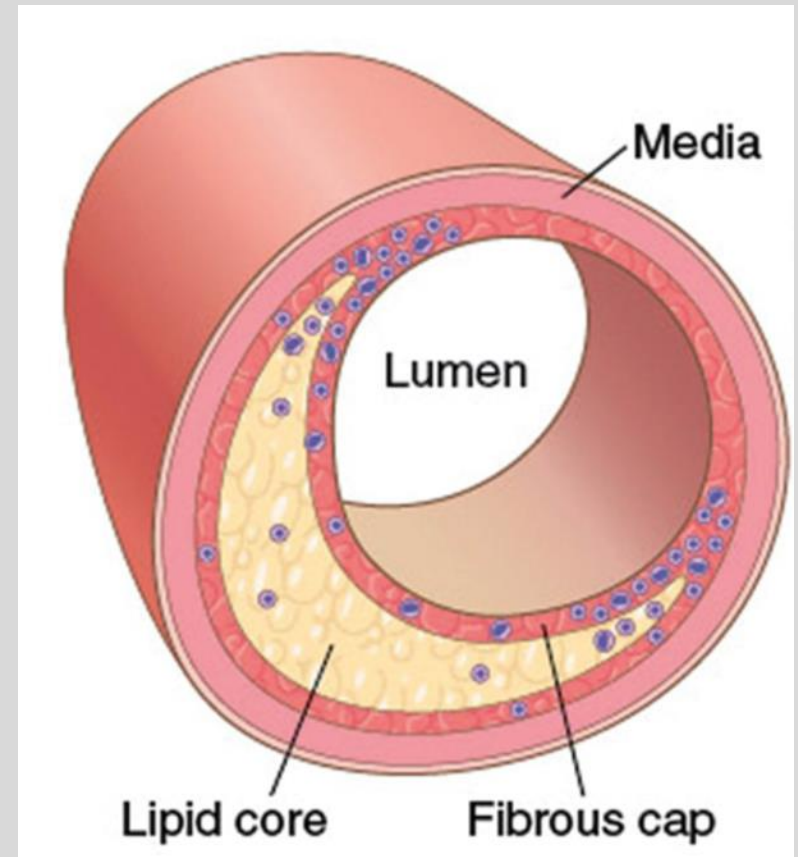
- **Stable Plaque**
 - Thick fibrous cap
 - Small lipid core
 - Minimal inflammation
 - Increased proliferation of smooth muscle cells
- Plaques rupture when they can no longer withstand the mechanical stress of shear forces



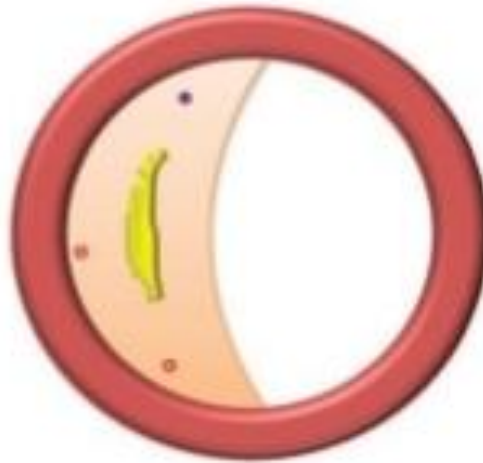


Unstable (Vulnerable) Plaques

- **Unstable (Vulnerable) Plaques**
 - Thin fibrous cap (few smooth muscle cells)
 - Large lipid core
 - High levels of inflammation (destabilizes the plaque)
 - Low levels of smooth muscle cells
 - **More likely to rupture**



Stable plaque



Small lipid core

Thick fibrous cap

Low macrophage content

Low microvessel density

No intraplaque hemorrhage

No cap rupture, no superimposed thrombus

Unstable, ruptured plaque



Large lipid core

Thin fibrous cap

High macrophage content

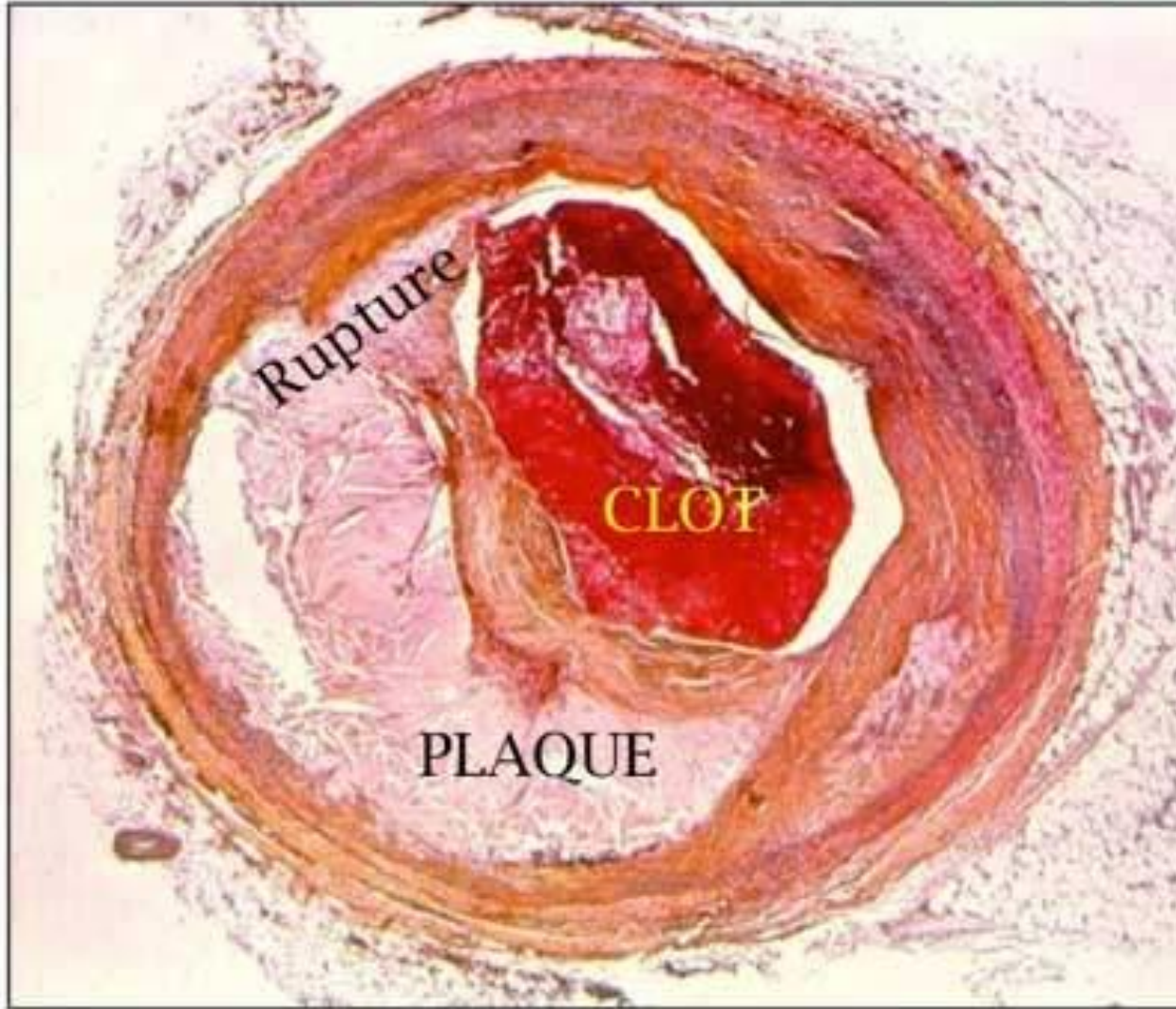
High microvessel density

Presence of intraplaque hemorrhage

Cap rupture and superimposed thrombus



The vulnerable atherosclerotic plaque



- large lipid core
- thin fibrous cap
- rich in macrophages
- increased MMPs
- poor in smooth muscle cells
- low-grade stenosis

Question

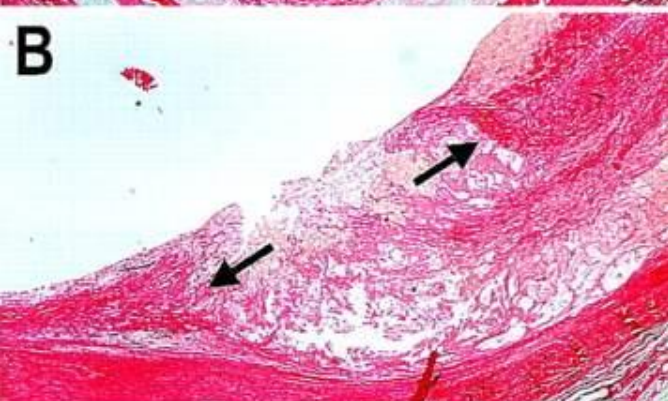
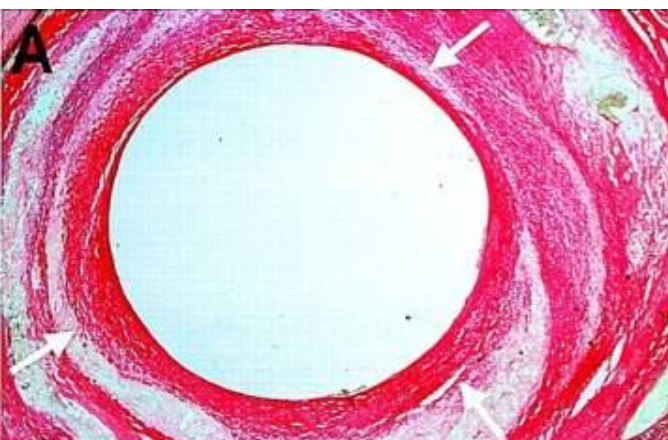
A 70-year-old male presents to the emergency department with slurred speech and left arm weakness that began 2 hours ago. He has a PMHx of hypertension. He denies currently smoking but mentions that he smoked 1 pack per day for 30 years. His medications include lisinopril and a multivitamin. On physical exam, the bruits are heard over the carotid arteries bilaterally. The physician suspects a stroke and immediate intervention to restore blood flow. The physician explains to the patient that the stroke was most likely caused by a clot that traveled from the carotid arteries and obstructed blood flow in the brain. Given the most likely cause, what microscopic findings would be expected to be found in this patient's carotid arteries?

- A. Thick fibrous cap
- B. Small lipid core
- C. Low levels of inflammation
- D. Low levels of smooth muscle
- E. 30% stenosis

Question

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- D. Low levels of smooth muscle**
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Acute Progression

- Plaque rupture or ulceration
- Prone to rupture when there is more inflammation
- Leads to thrombus formation which can embolize or locally obstruct blood flow
- End-organ damage (MI, Stroke, CKD)

Picro Sirius staining for collagen of atherosclerotic cross-sections.

(A) Atherosclerotic plaque with a fibrous cap overlying lipid-rich areas.

(B) Atherosclerotic plaque with the atheroma adjacent to the lumen due to rupture of the thin fibrous cap.

(C) Rupture of the fibrous cap

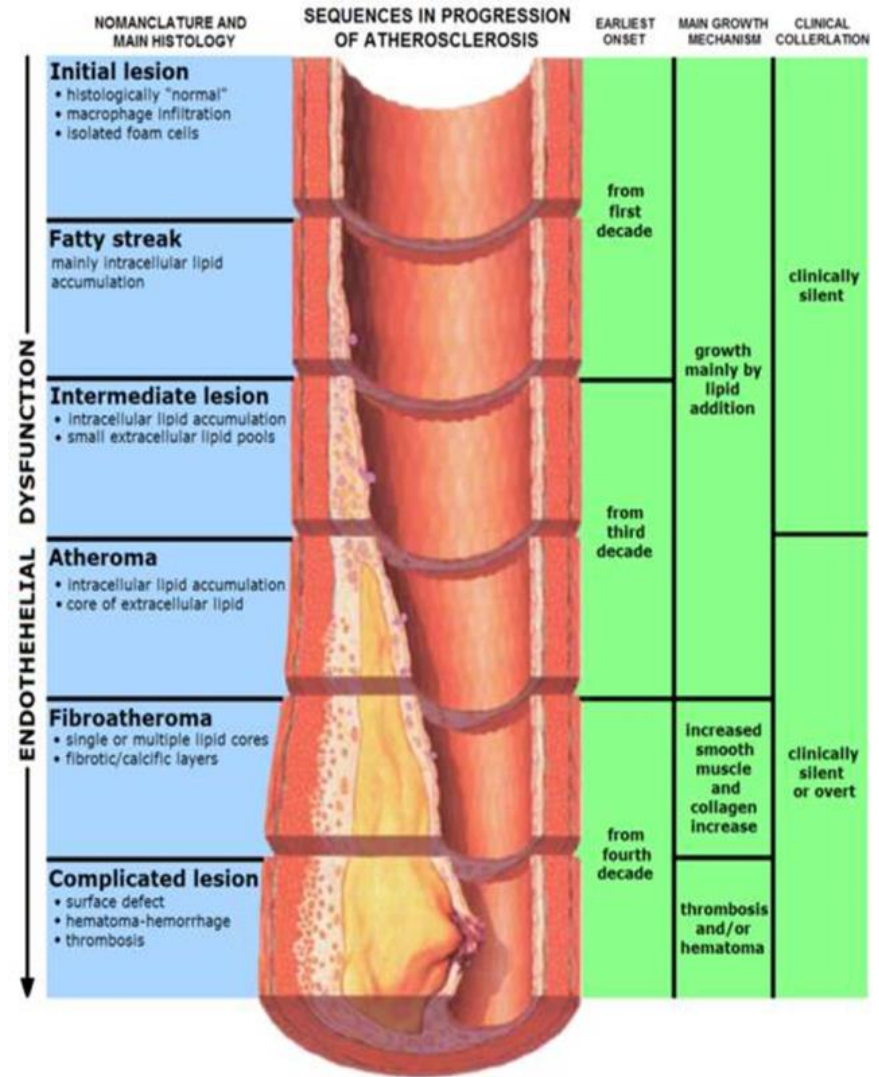
Pasterkamp, G. et al. J Am Coll Cardiol 2000;36:13-21



- **Chronic Progression**
 - Vessel narrowing
 - Obstructed/reduced blood flow
 - **Loss of elasticity**
 - **Cycle of damage**
 - Atherosclerosis promotes the development of HTN
 - HTN causes more endothelial damage
 - More damage leads to more plaque formation

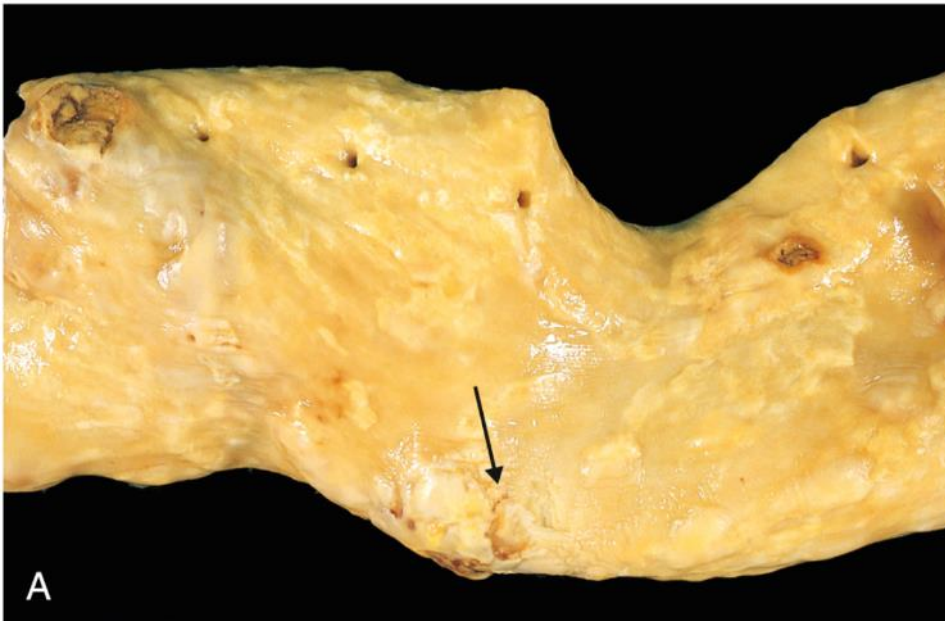
Atherosclerosis Progression

- Type I – isolated macrophage foam cells
- Type II – intracellular lipid accumulation
- Type III - Type II + small extracellular lipid pools
- Type IV – Type II + core of extracellular lipid
- Type V – lipid core and fibrotic layer, or multiple lipid cores, fibrotic/ calcific layer
- Type VI – surface defect; hematoma/ hemorrhage/ thrombosis





Fatty streaks are the earliest lesions seen with atherosclerosis in arteries. (WEBPATH)



Atherosclerotic disease of the aorta

A) Mild atherosclerosis

Arrow – fibrous plaques

B) Severe atherosclerosis with diffuse, complicated lesions

Open arrow – ulcerated plaque

Closed arrow – lesion with an overlying thrombus



Mild, moderate, and severe atherosclerosis (bottom to top) of abdominal aorta

Question

A 50 year old man has episodes of severe substernal chest pain every time he performs a task requiring moderate exercise.

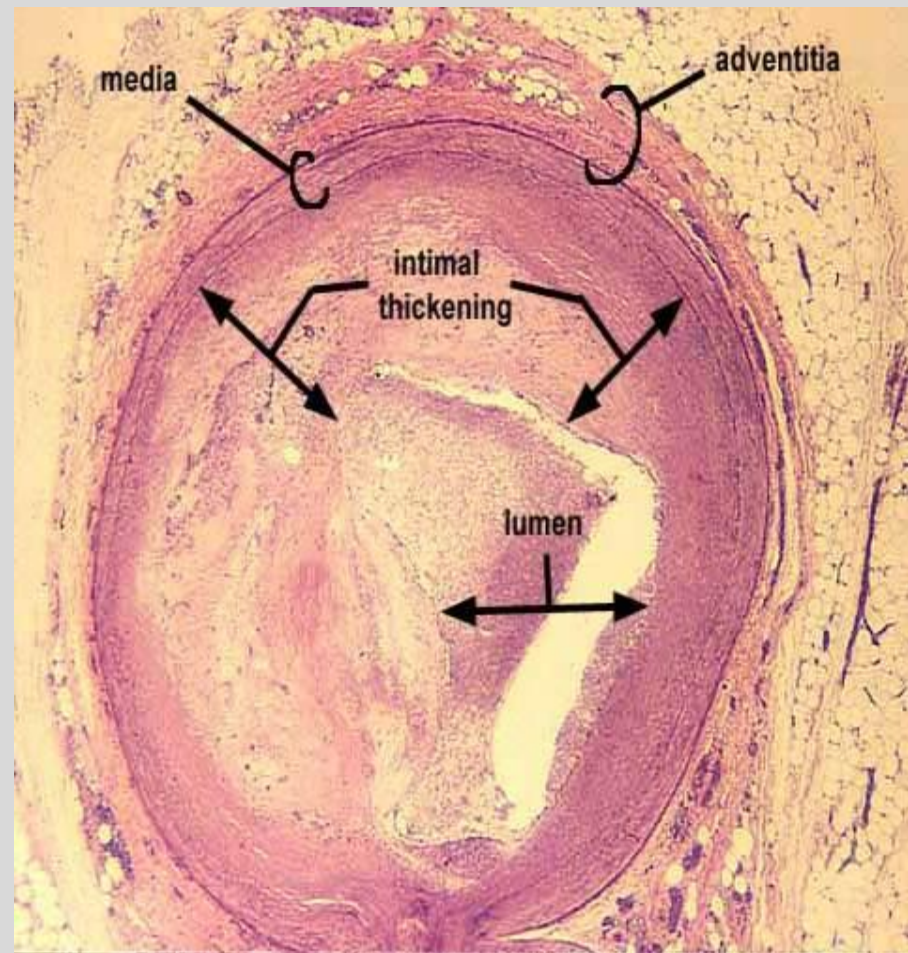
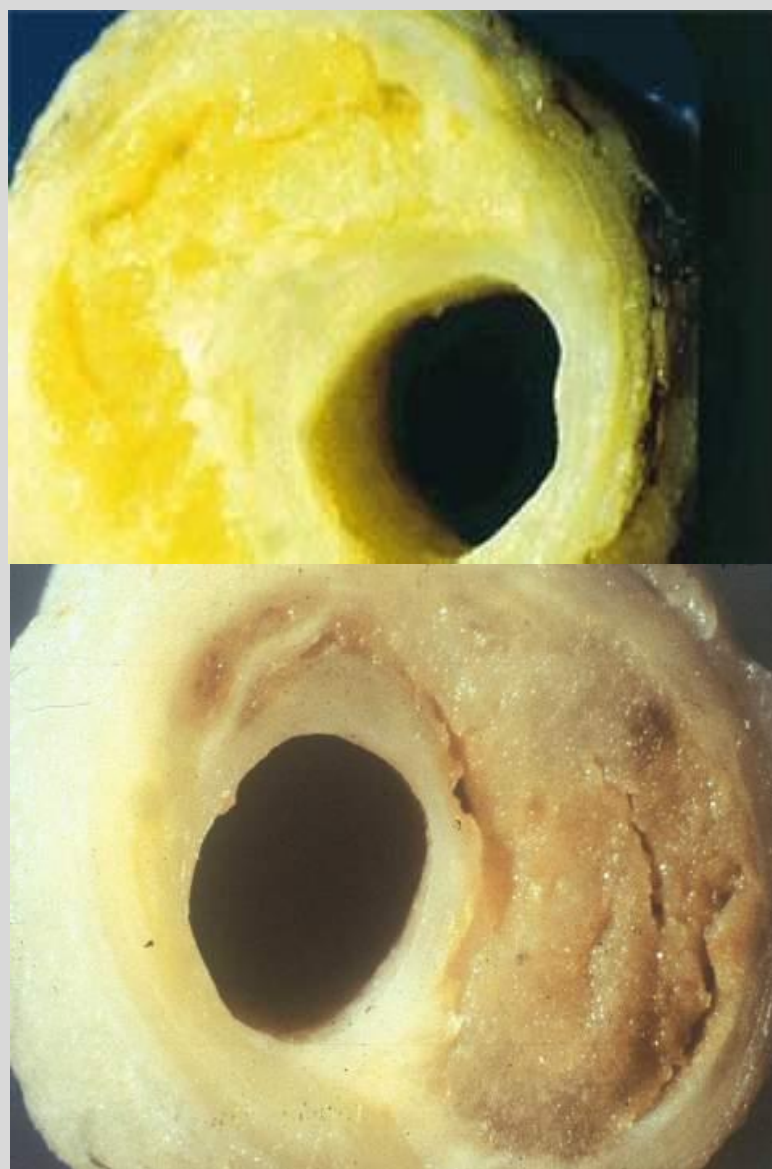
The episodes have become more frequent and severe over the past year, but can be relieved by nitroglycerin.

On PE, he is afebrile, has a heart rate of 78 bpm which is regular and has no murmurs or gallops.

His labs show a normal serum creatinine level, a mildly elevated serum glucose level and a mild to moderately elevated serum cholesterol level.

Which of the following cardiac lesions is most likely to be present at this time?

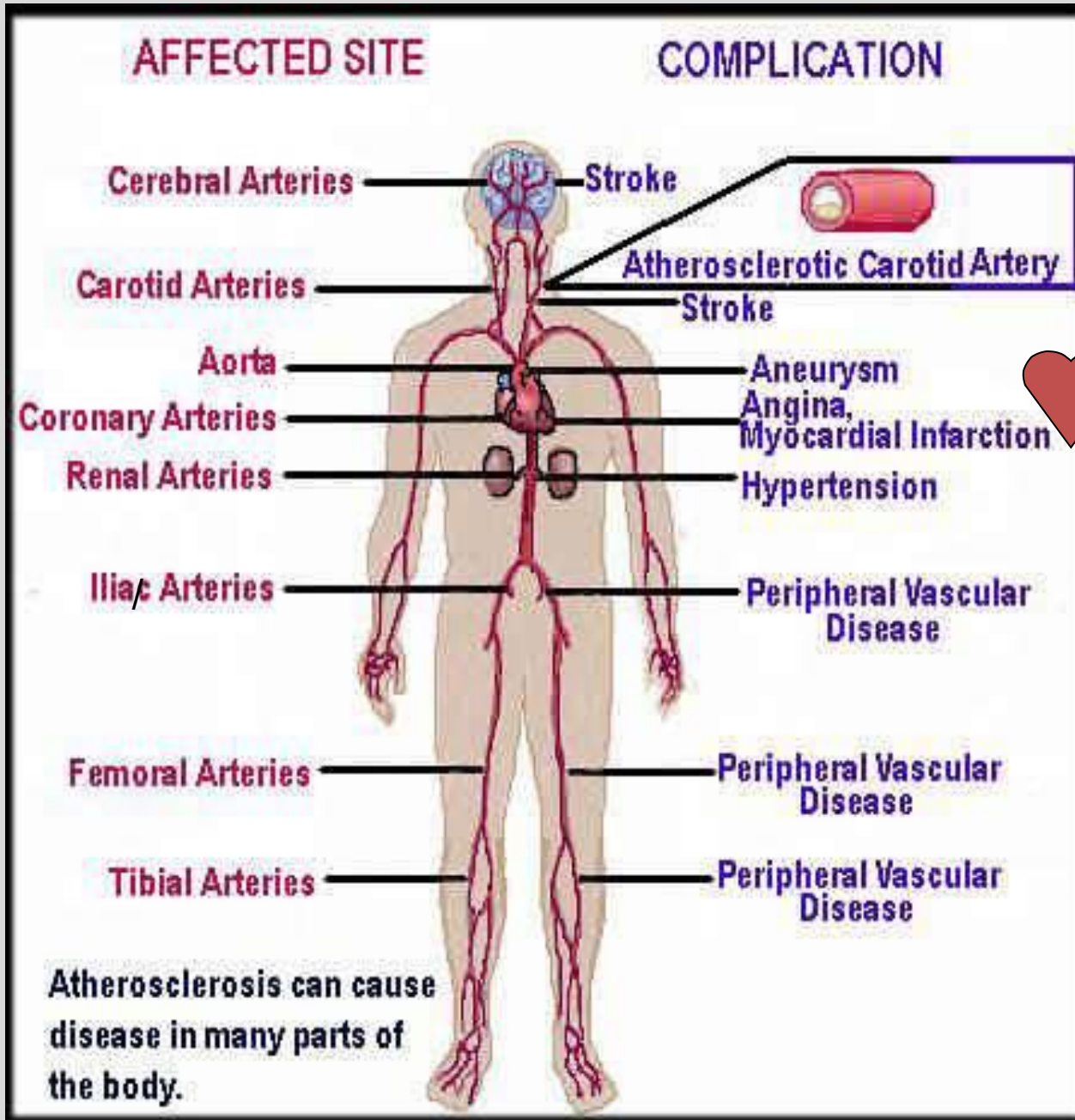
- A. Acute myocardial infarction
- B. Coronary atherosclerosis
- C. Calcific aortic stenosis
- D. Viral myocarditis
- E. Ruptured papillary muscle



- **Rupture, ulceration, erosion**
 - Exposes blood stream to highly thrombogenic substances
 - Leads to thrombosis and possible complete occlusion
 - **Thrombosis can lead to embolism** (atheroembolism) and eventual end-organ ischemia
- **Hemorrhage into plaque**
 - May extend the plaque or induce rupture

- **Aneurysm**

- Outpouching of the vessel wall due to weakness
- **Thickened intima means less O₂ diffusion to the rest of the vessel wall**
- **Media and adventitia layers become atrophic**
 - Atrophy causes weakening of the walls and eventual aneurysm

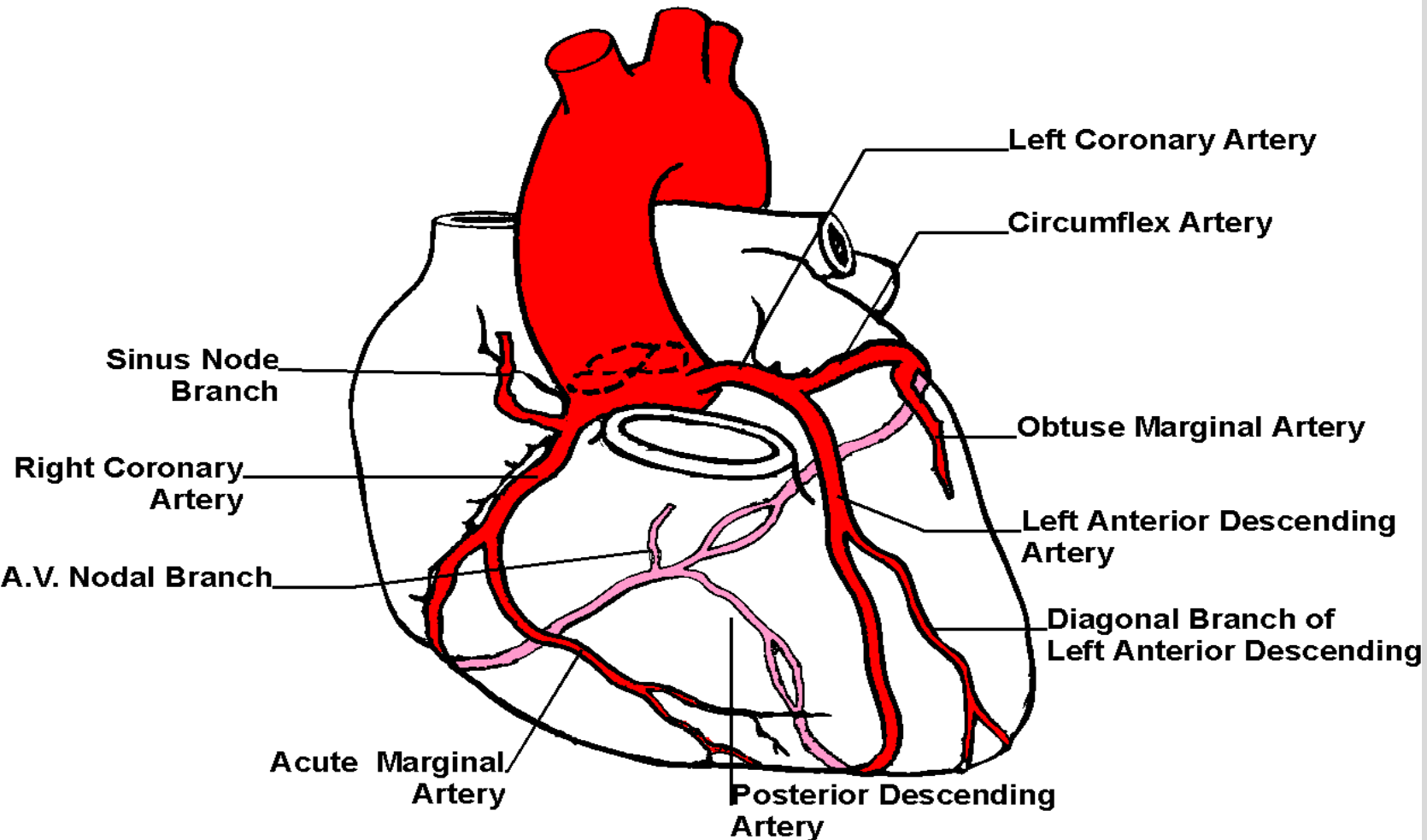


Clinical Presentation

- Develops slowly and usually **asymptomatic until >70-75% stenosis**, thrombosis or embolization occurs
- Organs that can be affected:
 - **Heart** – angina, MI (coronary arteries)
 - **Brain** – TIA, stroke (carotid arteries)
 - **Eyes** – retinal artery/vein occlusion
 - **Extremities** – PVD, gangrene, acute limb ischemia
 - **Kidney** - renal artery stenosis, CKD
 - **Intestine** – chronic mesenteric ischemia (intestinal angina)

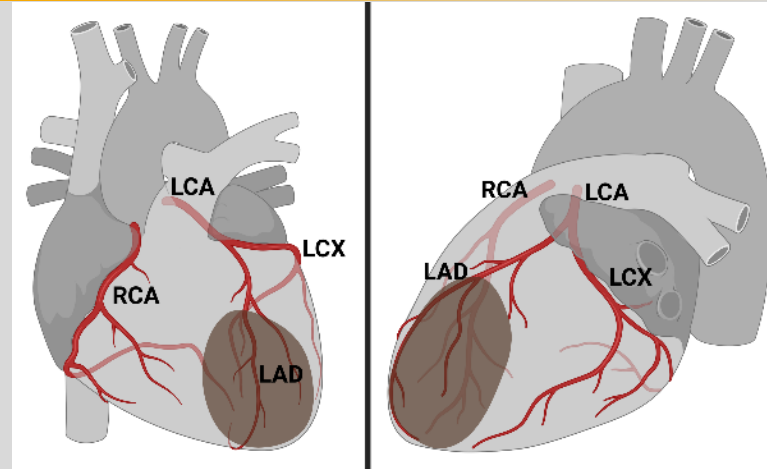
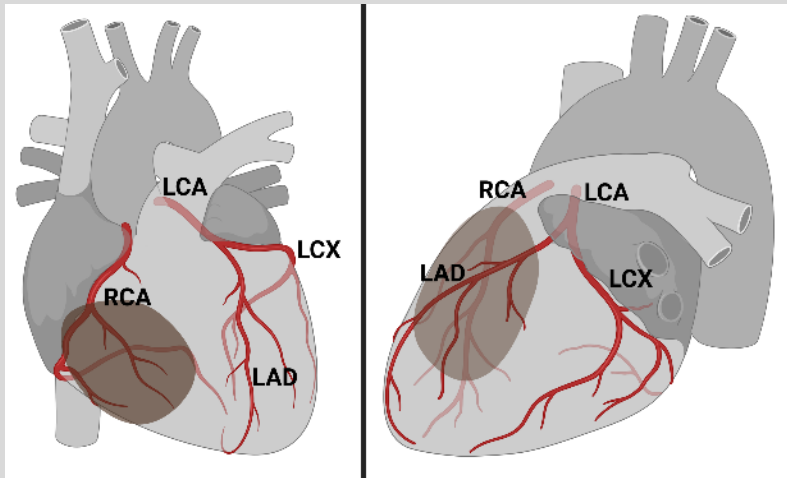
Blood Supply: Coronary Arteries

Coronary Arteries



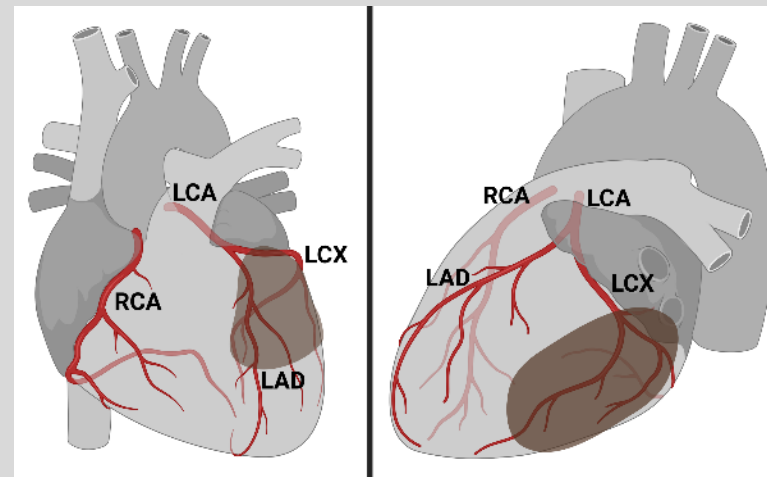
Blood Supply of the Heart - REVIEW

- **Left Anterior Descending:** apex of heart, anterior wall of the left ventricle, and anterior two-thirds of the ventricular septum



- **Right Coronary Artery:** entire right ventricle wall, posterior-basal (inferior) wall of the left ventricle, and posterior third of ventricular septum

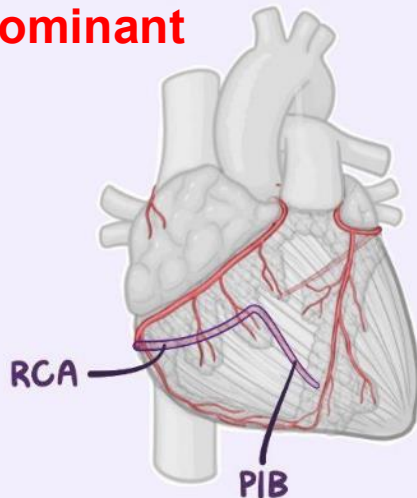
- **Left Circumflex:** only perfuses the lateral wall of the left ventricle



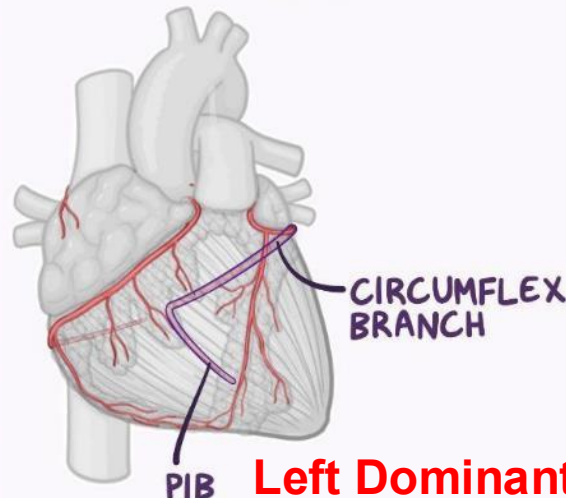
Blood Supply of the Heart - REVIEW

- The **coronary artery that gives rise to the posterior descending (posterior interventricular) branch** is considered the **dominant side**
 - Perfuses posterior third of septum
 - Most people **are right dominant** (RCA gives rise to the PIB)

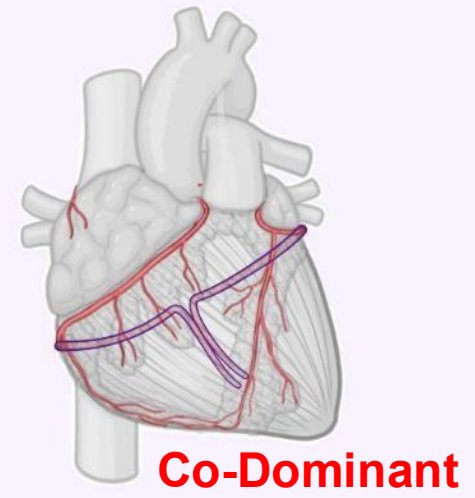
Right 67-85%
Dominant



8-15%

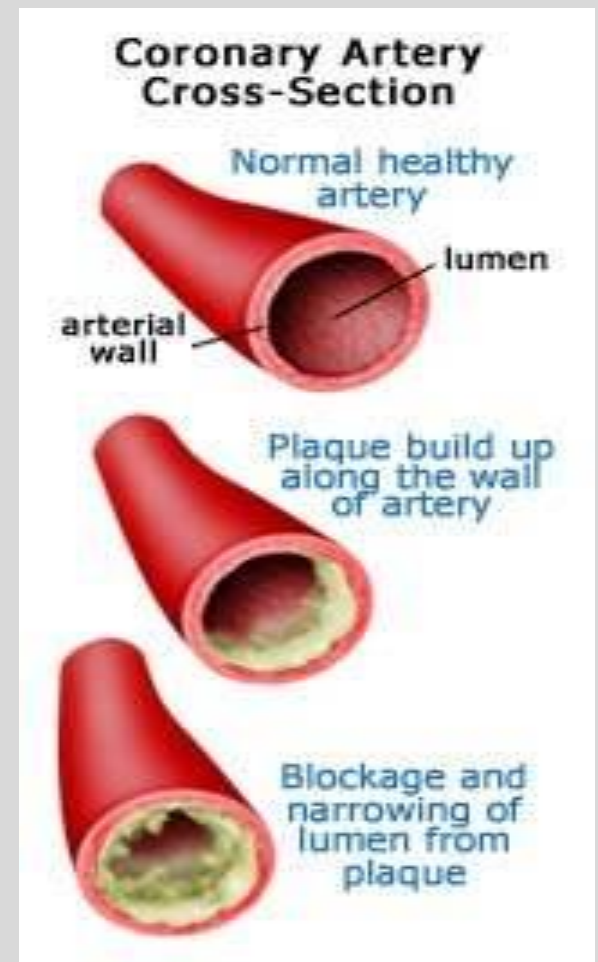
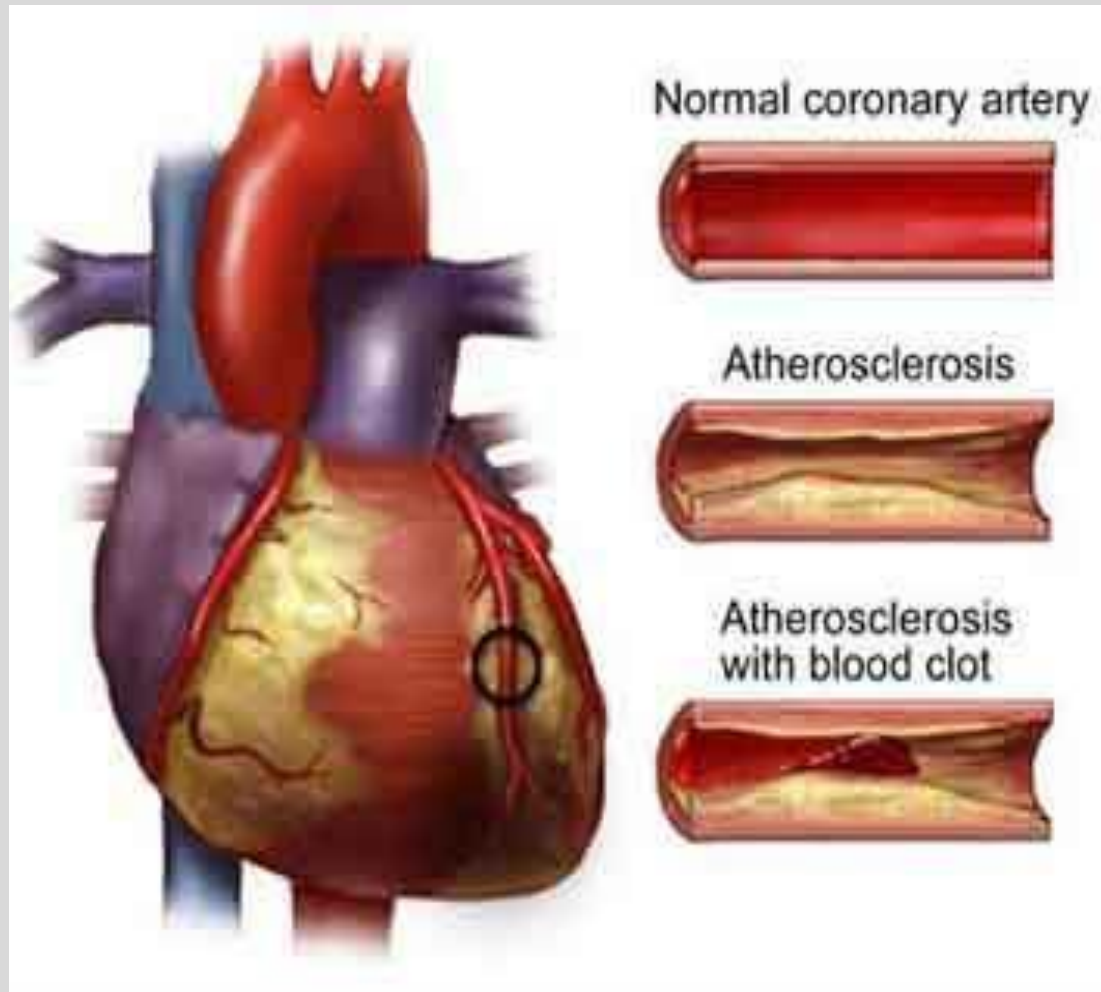


7-18%



Coronary Artery Disease

- Most patients with ischemic heart disease have atherosclerosis of one or more coronary arteries
- Either progressive stenosis of lumen (chronic, “fixed” obstruction) or acute plaque disruption with thrombosis (sudden, dynamic).
- 70 - 75 % stenosis causes symptomatic ischemia induced by exercise (stable angina); 90 % stenosis leads to inadequate coronary blood flow at rest



Question

A 47 year old man has chest pain upon exertion. He was well until four months ago when he first developed chest discomfort while jogging.

His symptoms have progressed so that now he has chest pain after climbing one flight of stairs. He has a history of diabetes controlled by diet and has smoked one pack of cigarettes per day for 25 years.

His father and maternal grandfather both died of heart disease before age 60. Because of the patient's chest pain, he is hospitalized, and on the 5th hospital day, he develops chest pain with mild activity which is minimally responsive to sublingual nitroglycerin.

Laboratory studies show a serum CK-MB of 4.5 ng/ml (N = 0 – 5 ng/ml) and a serum troponin-I of 0.3 ng/ml (N < 0.4 ng/ml).

Which is the most likely diagnosis?

- A. Acute myocardial infarction
- B. Cardiac arrhythmia
- C. Dressler syndrome
- D. Stable angina
- E. unstable angina

Ischemic Heart Disease

- Myocardial ischemia – imbalance between supply and demand heart for oxygenated blood
- Reduction in coronary blood flow secondary to atherosclerotic coronary arterial obstruction (coronary artery disease)

Clinical manifestations:

- **Angina pectoris: less severe ischemia, no death of cardiac muscle**
- **Myocardial infarction: death (necrosis) of cardiac muscle**
- Chronic ischemic heart disease with heart failure
- Sudden death

ANGINA PECTORIS - Sudden, recurrent attacks of substernal/ precordial chest pain caused by transient (15 seconds to 15 minutes) myocardial ischemia; no necrosis/infarct

Stable (typical) angina:

- reduction of coronary perfusion secondary to marked chronic stenosis of coronary artery
- usually not associated with plaque disruption
- relieved by rest or nitroglycerin (vasodilator)

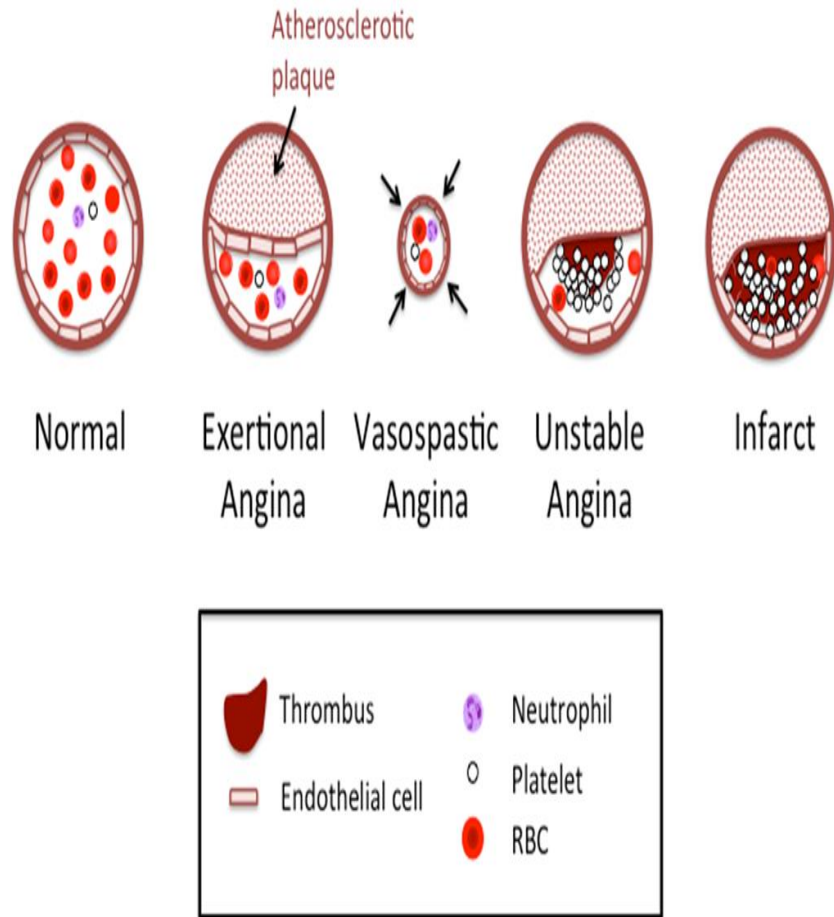
Unstable angina:

- increased progressive frequency, usually at rest, more prolonged
- disruption of AS plaque with superimposed partial thrombosis or vasospasm (preinfarction angina)

Prinzmetal angina:

- occurs at rest due to coronary artery spasm; uncommon
- unrelated to exercise, heart rate, or bp
- relieved by nitroglycerin or calcium channel blockers

ANGINA



- **Stable (exertional) angina** - arterial lumen narrowed by stable plaque: partial stenosis
- **Vasospastic angina (Prinzmetal):** Inappropriate vasoconstriction, intense vasospasm, usually in absence of plaque
- **Unstable angina:** plaque rupture results in platelet aggregation and formation of a thrombus, **NON-occlusive**
- Major coronary artery completely obstructed by thrombus: infarction

• Adapted from Rhee et al (2011).

Syndrome	Description
Angina Pectoris	Uncomfortable sensation in the chest and neighboring anatomic structures produced by myocardial ischemia
Stable angina	Chronic pattern of transient angina precipitated by physical activity or emotional upset, relieved by rest within a few minutes. Oxygen demand exceeds available blood supply. Episodes are associated with temporary depression of ST segment but permanent myocardial damage does not result.
Unstable Angina	Pattern of increased frequency and duration of angina episodes produced by less exertion or at rest. Oxygen demand is unchanged, supply is decreased secondary to reduced resting coronary flow.
Variant(Prinzmetal) angina	Typical angina discomfort, usually at rest, which develops because of coronary artery spasm rather than an increase of myocardial oxygen demand.

Question

A 68 year old obese woman presents with substernal chest pain and a history of recurrent angina pectoris and intermittent claudication.

Laboratory studies show a serum CK-MB of 6.6 ng/ml (N = 0 – 5 ng/ml) and a serum troponin-I of 2.5 ng/ml (N < 0.4 ng/ml).

EKG changes showing ST elevations are also present. Which of the following mechanisms is most likely responsible for these results?

- A. Coronary artery thrombosis
- B. Coronary artery vasospasm
- C. Decreased collateral blood flow
- D. Deep venous thrombosis
- E. Paradoxical embolism

Acute coronary syndromes

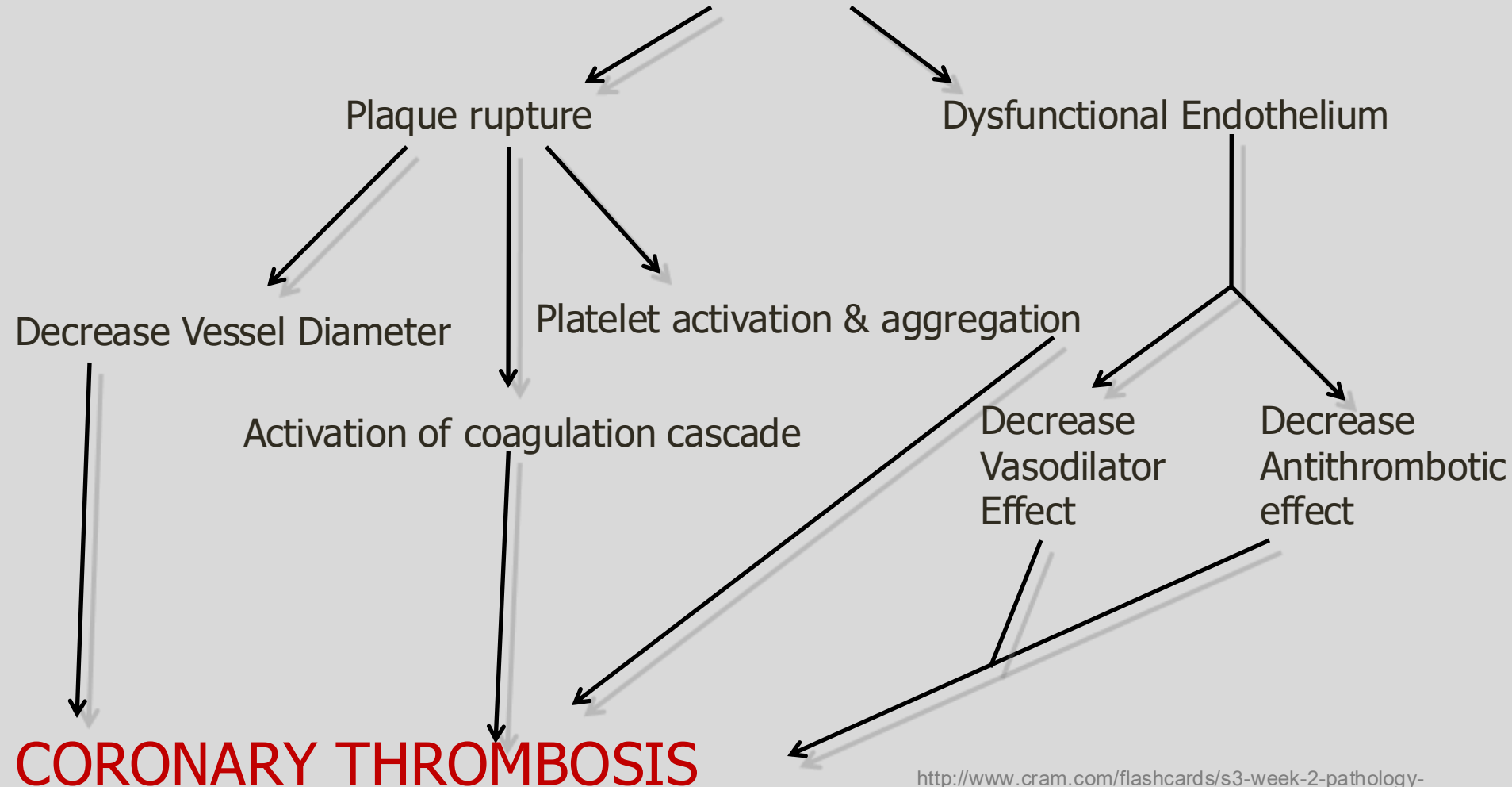
- **Unstable angina, acute MI, and sudden cardiac death**
- Usually precipitated by abrupt atherosclerotic plaque change/rupture followed by thrombosis
- Disruption of previously partially stenotic plaques by
 - rupture - exposes thrombogenic plaque components
 - erosion/ulceration - exposes thrombogenic subendothelial basement membrane
 - hemorrhage into atheroma
- Vasoconstriction may play role by constricting lumen, narrowing size and promoting plaque disruption
- Vasoconstriction stimulated by factors such as circulating adrenergic agonist, locally released platelet contents, etc.

Coronary thrombosis

- Most acute coronary syndromes associated with disruption of the AS vulnerable plaque (not fixed, stable plaque)
- **MI often occurs when thrombus is superimposed on disrupted (previously partially stenotic) plaque, transforming it into a total occlusion**
- Many ruptured plaques with subsequent coronary thrombosis resulting in MI occur in coronary arteries which were previously only partially stenotic (many less than 70% stenosis)

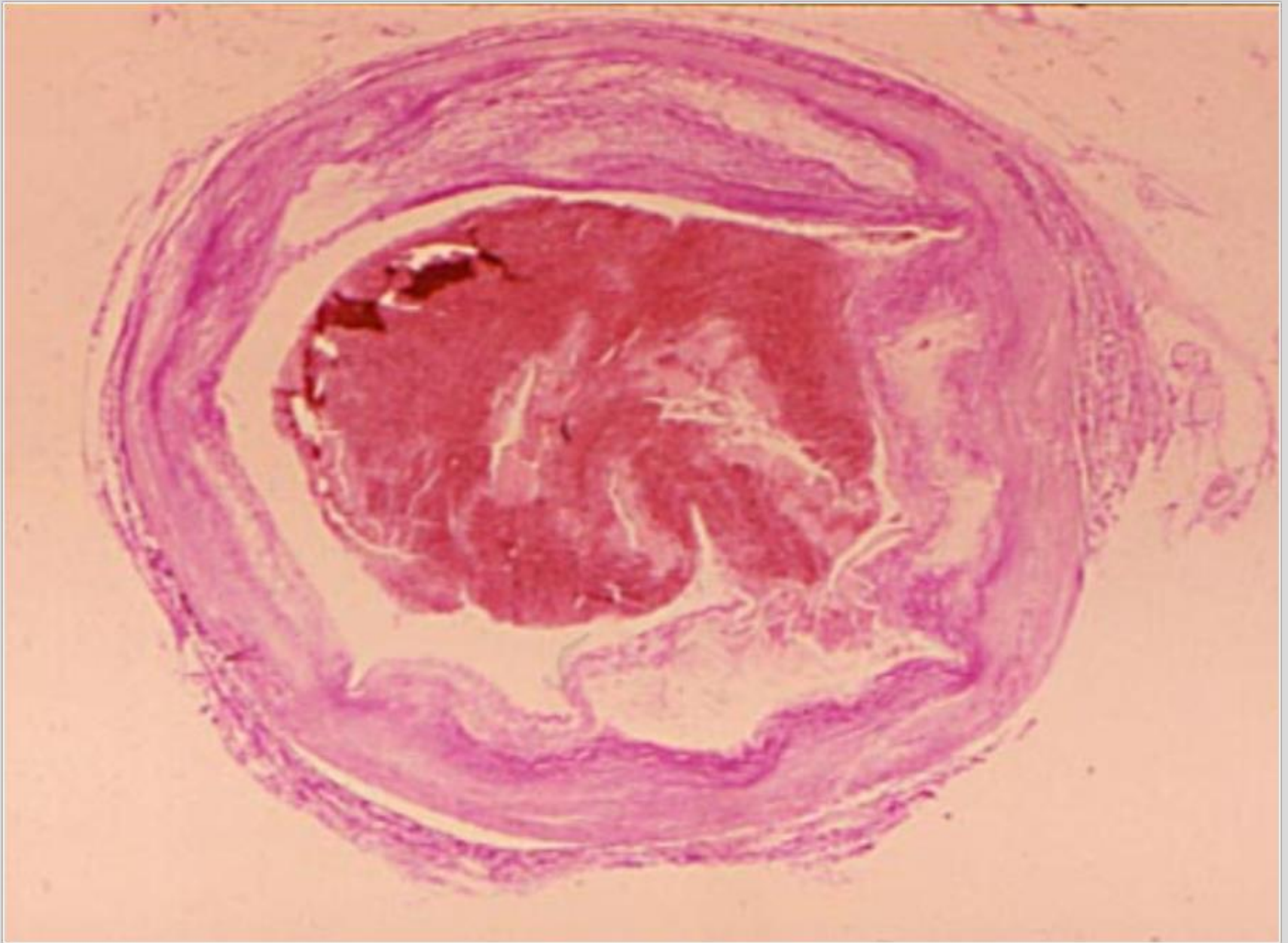
Mechanism of Coronary Thrombosis (Acute Coronary Syndromes)

Atherosclerosis



Coronary atherosclerosis with hemorrhage into atheromatous plaque.





Summary Slide

- Atherosclerosis/atheroma
- Risk factors
- Atherogenesis
 - Endothelial injury
 - Role of hyperlipidemia, hemodynamic effects, etc.
 - Accumulation of lipids
 - Inflammation – wbc's – macrophages – foam cells
 - Migration of smooth muscle cells from media to intima, proliferation, ECM elaboration
 - Growth and maturation of plaque
- Progression of atherosclerosis
- Complications of atherosclerosis – rupture, ulceration, aneurysm, thrombosis
- Vulnerable v. stable plaques
- Coronary artery diseases – atherosclerosis involving coronary arteries – ischemic syndromes
- Stable, unstable, and Prinzmetal angina
- Rupture of partially stenotic vulnerable plaque with superimposed thrombosis – coronary thrombosis leading to total lumen occlusion and myocardial infarction

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

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